

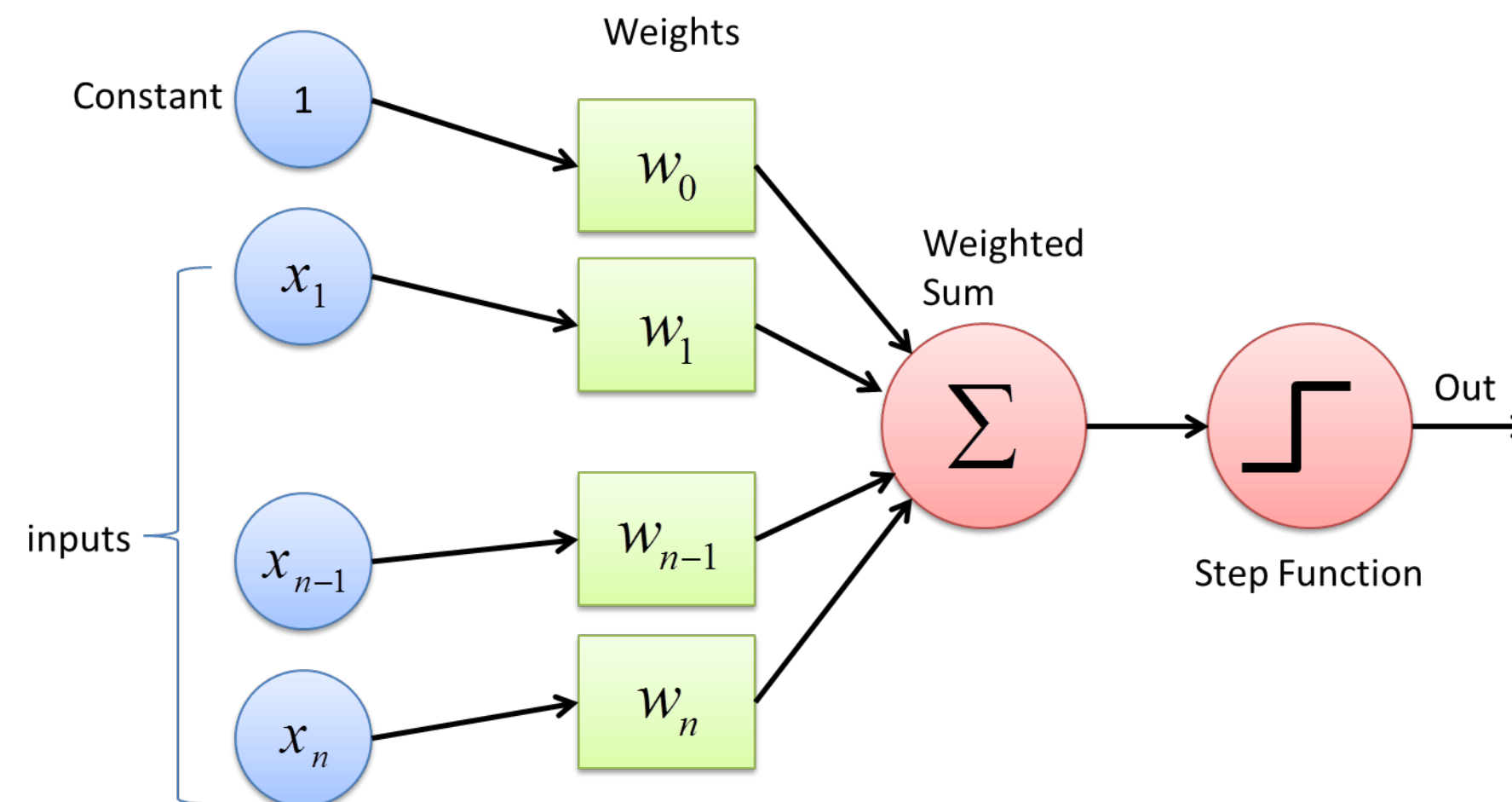
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# **An Introduction to Multi-Modality**

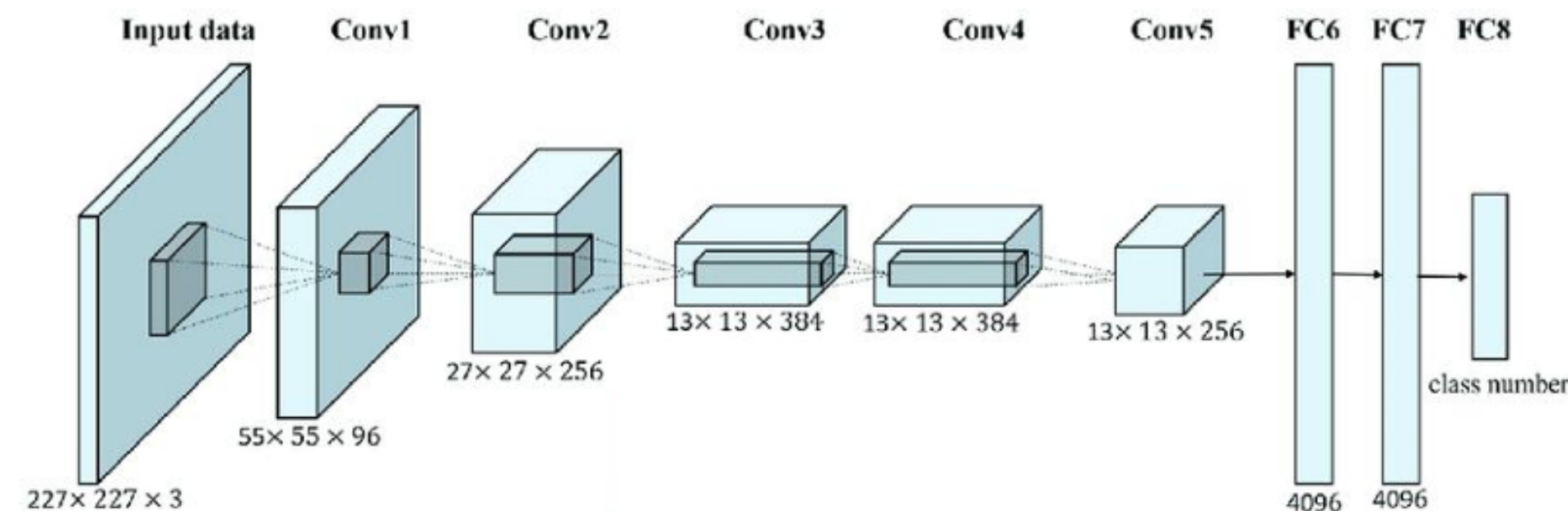
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- ▶ **Attention/Neural Networks**
- ▶ **Multi-Modality and Fusion Methods**
- ▶ **Multi-Modal Applications**

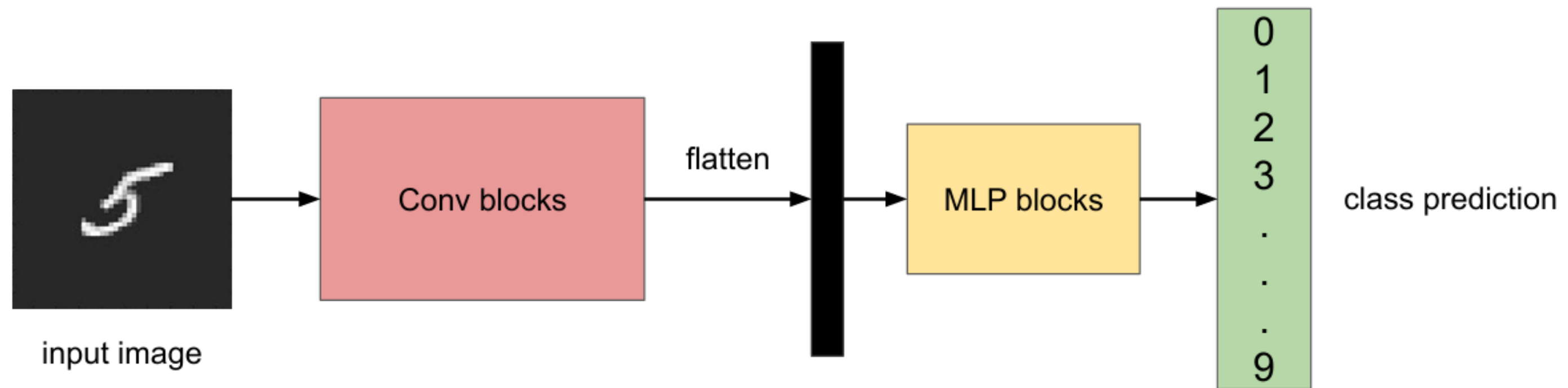
- ▶ **Neural Networks consist of multiple building units called neurons, meant to represent the way the human brain functions.**
- ▶ **Inputs are weighted, added and then passed through a function that produces the output**
- ▶ **The output is utilized for the task at hand, whether it is classification or segmentation, for example.**



- ▶ A classic model for image tasks are **Convolutional Neural Networks**
- ▶ Through each layer, a convolutional filter is utilized in order to shrink the image into an embedding space, a representation of information
- ▶ The representation is used to make decisions regarding classification and segmentation





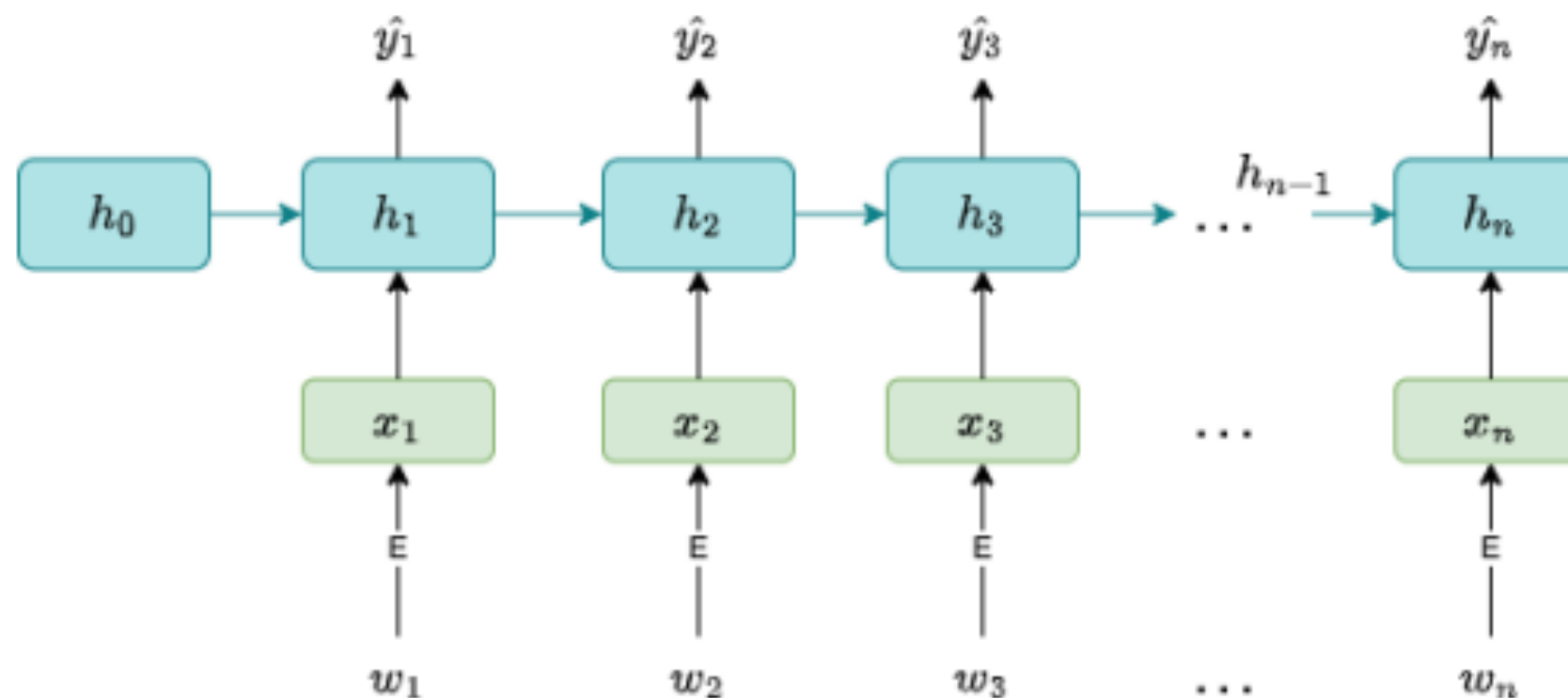


## Now, we are going to focus on sequential data

- ▶ The main form of sequential data that artificial intelligence is utilized upon is text
- ▶ Text is basically a combination of various words that provides a certain meaning
- ▶ There is difficulty in containing the information about words thoroughly through a model

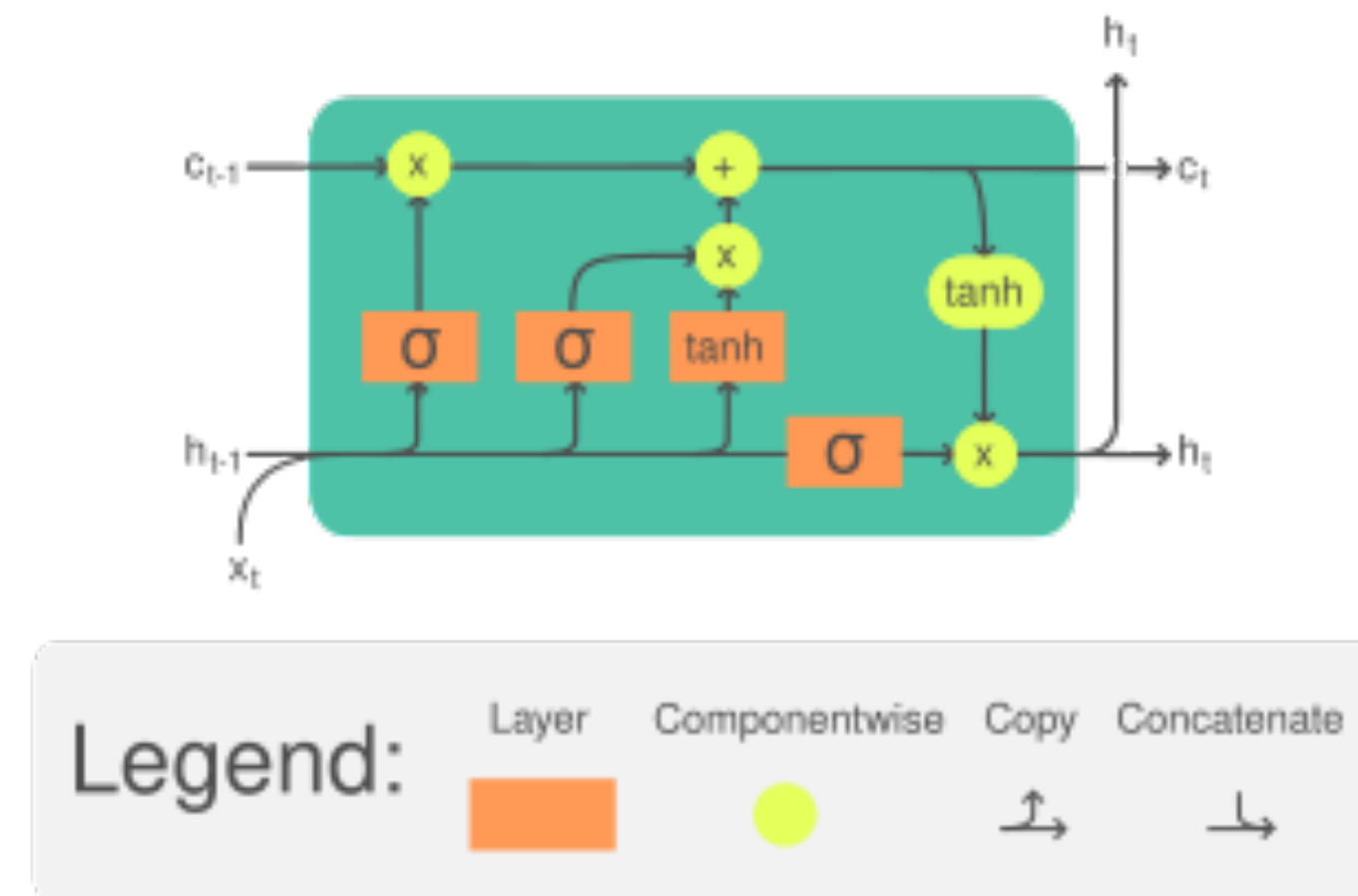
The first attempt is with Recurrent Neural Networks.

- ▶ RNNs utilize recurrent connections, where the output of a neuron at one time step is fed back as input to the network at the next time step
- ▶ Enables capturing of temporal dependencies and patterns within sequences



## Drawbacks:

- ▶ **Vanishing Gradient:** As we move forward, the gradient becomes smaller
- ▶ **As a result, we cannot train when the sequences are very large**
- ▶ **Long Short Term Memory Models (LSTMs) and Gated Recurrent Units (GRU) were implemented**
- ▶ **Suffer from increased complexity and risks for overfitting**



## Attention arrives to solve our problems:

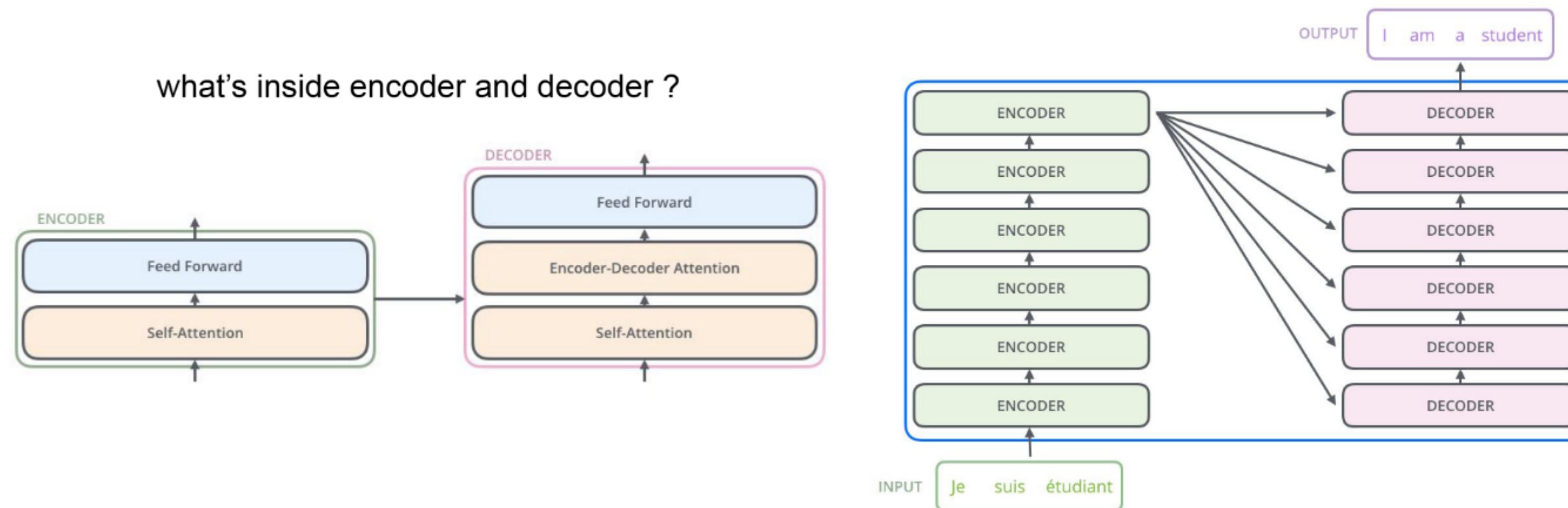
- ▶ **Attention determines the relative importance of each component in a sequence relative to the other components**
- ▶ **For text, this is extremely important as we leverage knowledge from all components to determine the meaning**
- ▶ **Query (Q): Represents the element that is "asking" for context. Acts as the search query in the attention process.**
- ▶ **Key (K): Encodes features for each element in the input sequence. Helps in matching the query with the relevant parts.**
- ▶ **Value (V): Contains the actual information or content of each element. Gets weighted and summed based on the query-key similarity.**

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax} \left( \frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}} + \mathbf{M} \right) \mathbf{V}$$



## Encoder-Decoder:

- ▶ **Encoder:** Encodes the information into a latent space of lower dimensionality
- ▶ **Decoder:** Decodes the information appropriately for the task, i.e. a mask for image segmentation, class probabilities for classification
- ▶ **We can combine attention in the encoder-decoder sections of a neural network model**

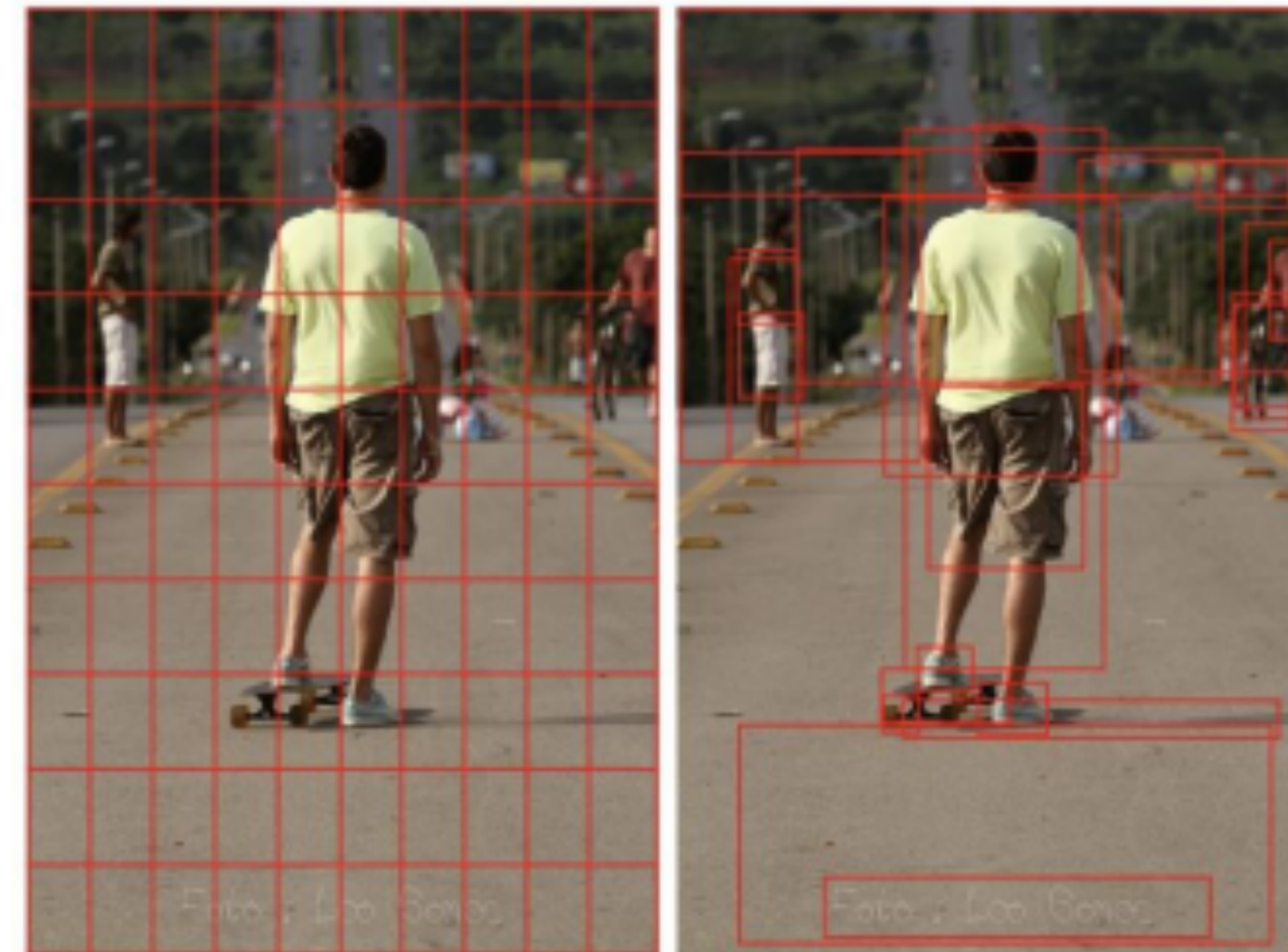


# Features

Featurizing text:  $\text{Batch\_size} \times \text{Sequence\_length} \times \text{Hidden\_size}$ .

Featurizing images:

- Sparse “region” features:
  - Object detectors
- Dense features:
  - ConvNet layer(s) or feature maps
  - Vision Transformer layers

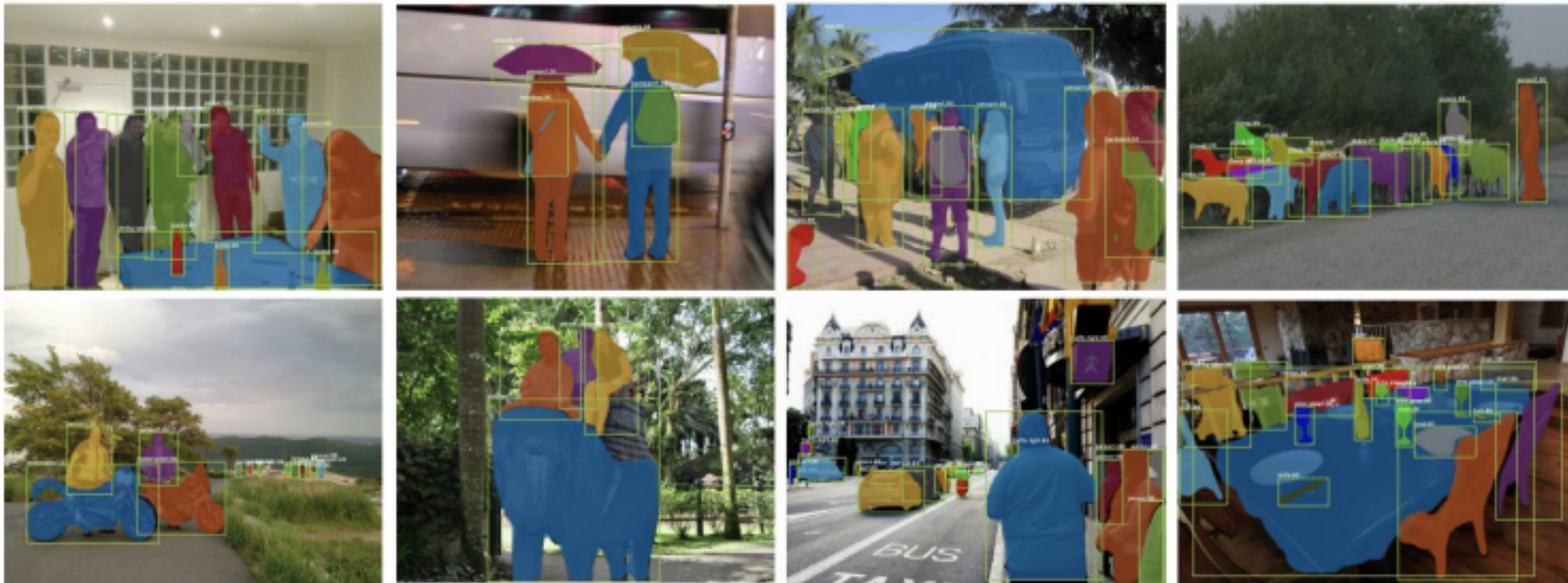


Anderson et al., 2018



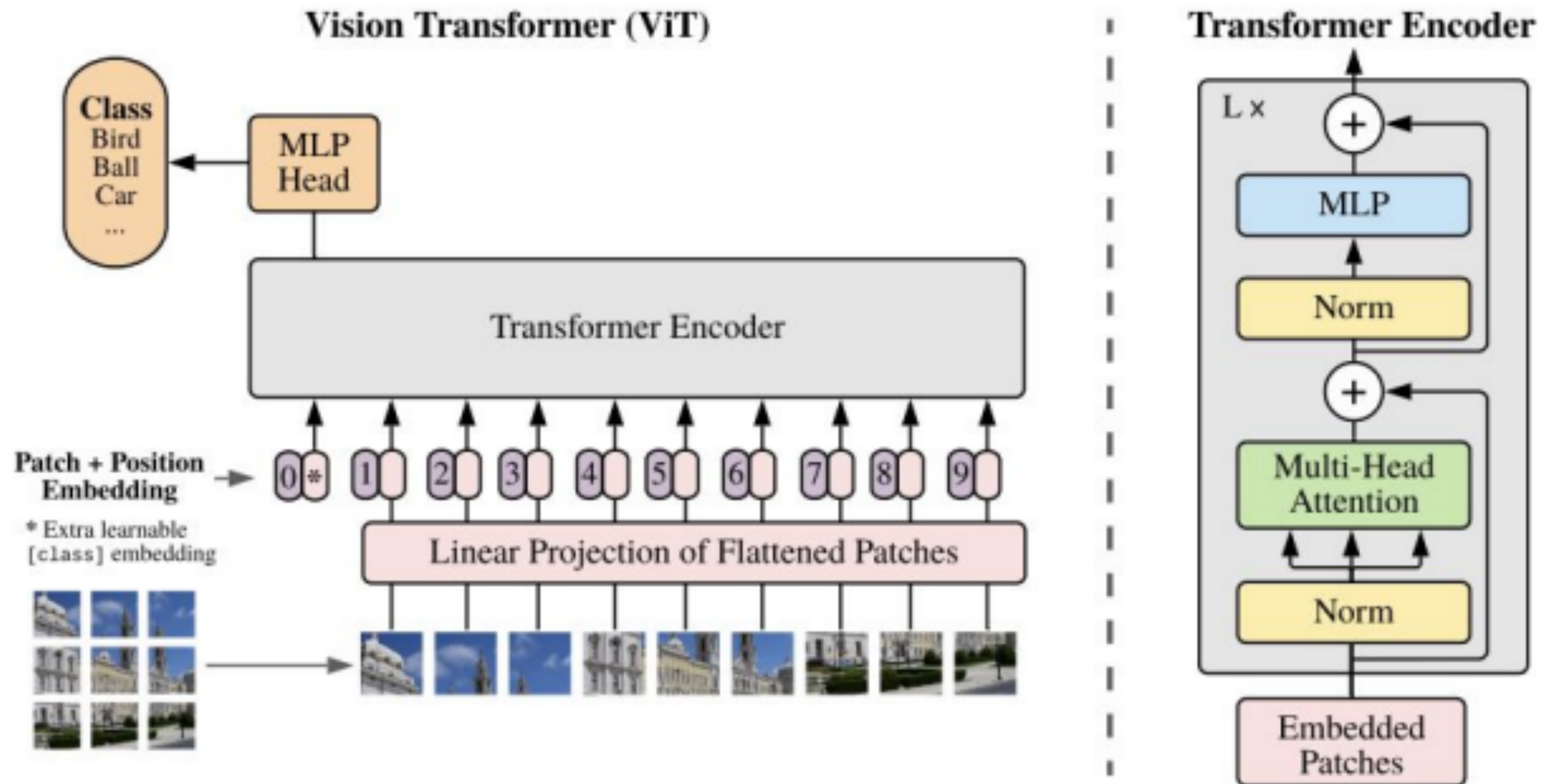
## Region features

- R-CNN (Girshick et al., 2014); Fast R-CNN (Girshick, 2015); Faster R-CNN (Ren et al., 2015); YOLO (you only look once) vX..





# Vision Transformers (Dosovitskiy et al., 2020)







# Attention as visual-semantic alignment



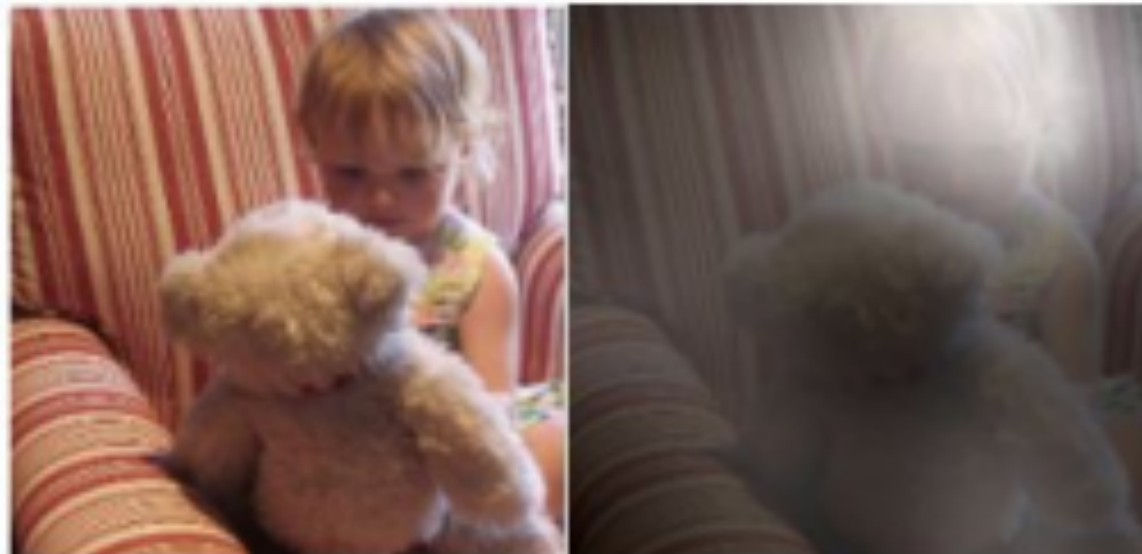
A woman is throwing a frisbee in a park.



A dog is standing on a hardwood floor.



A stop sign is on a road with a mountain in the background.



A little girl sitting on a bed with a teddy bear.



A group of people sitting on a boat in the water.



A giraffe standing in a forest with trees in the background.

# Problems with multimodality

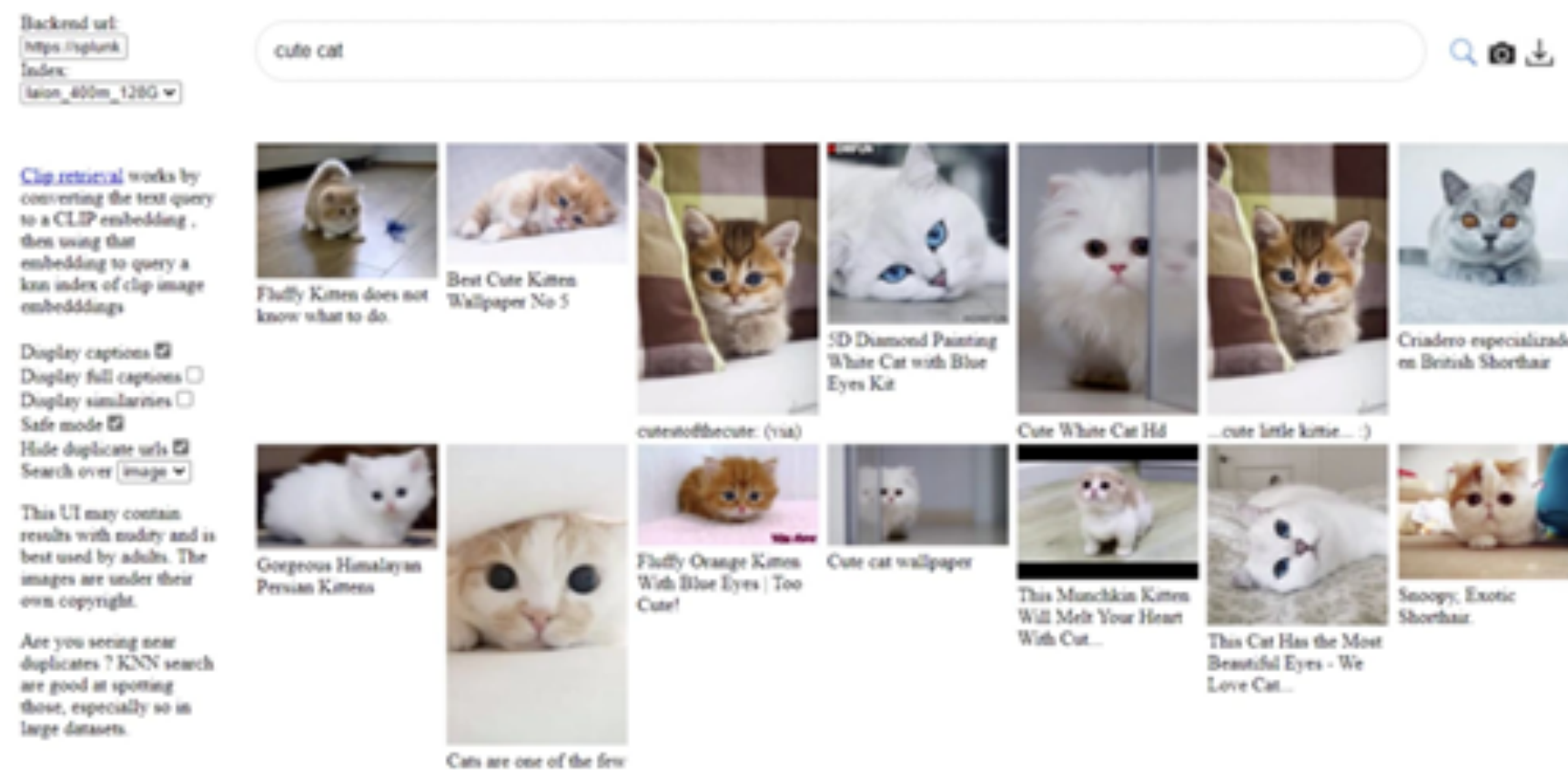
If it's so important, why isn't every system multimodal from first principles?

- One modality can dominate other modalities.
- Additional modalities can add noise.
- Full coverage over modalities is not guaranteed.



## Example: LAION-Dataset

- ▶ Dataset consisting of image and text pairs
- ▶ 2 Versions: 400M and 5B pairs





# Multimodal fusion

## Similarity

- Inner product:  $\mathbf{u}\mathbf{v}$

## Linear / sum

- Concat:  $W[\mathbf{u}, \mathbf{v}]$
- Sum:  $W\mathbf{u} + V\mathbf{v}$
- Max:  $\max(W\mathbf{u}, V\mathbf{v})$

## Multiplicative

- Multiplicative:  $W\mathbf{u} \odot V\mathbf{v}$
- Gating:  $\sigma(W\mathbf{u}) \odot V\mathbf{v}$
- LSTM-style:  $\tanh(W\mathbf{u}) \odot V\mathbf{v}$

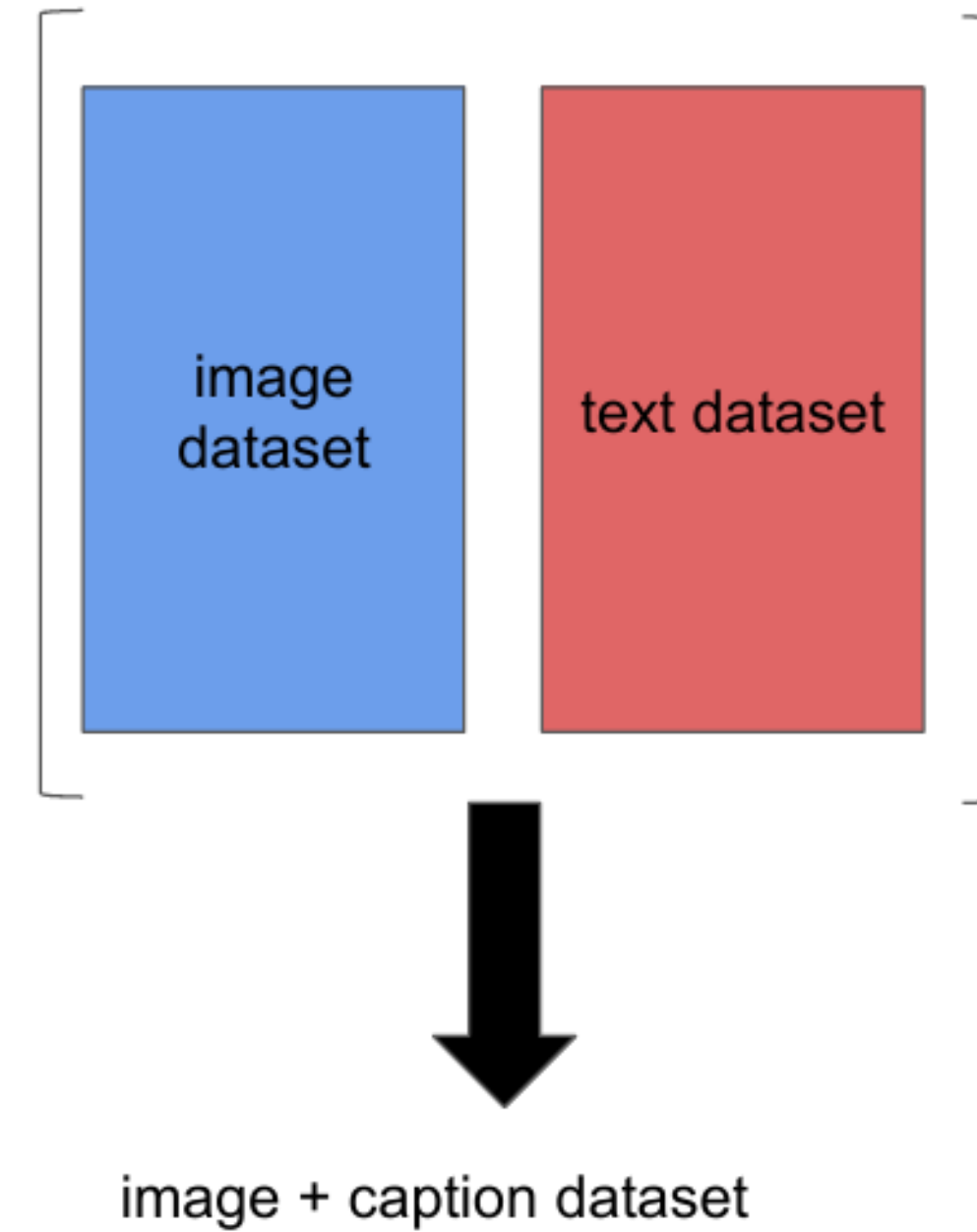
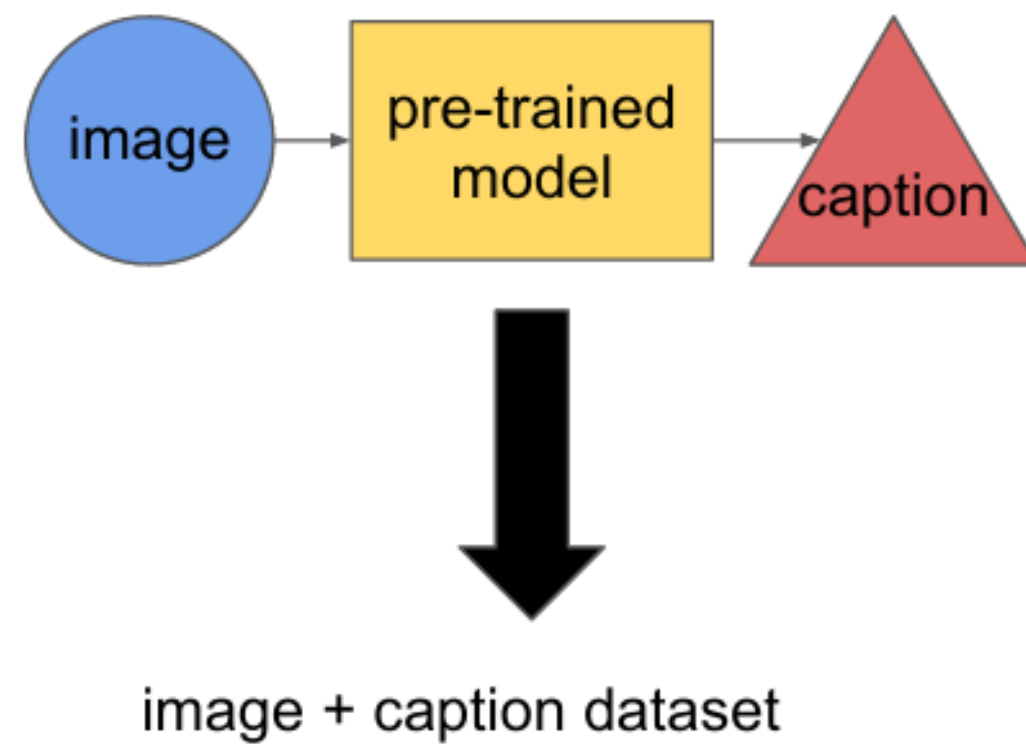
## Attention

- Attention:  $\alpha W\mathbf{u} + \beta V\mathbf{v}$
- Modulation:  $[\alpha\mathbf{u}, (1-\alpha)\mathbf{v}]$

## Bilinear

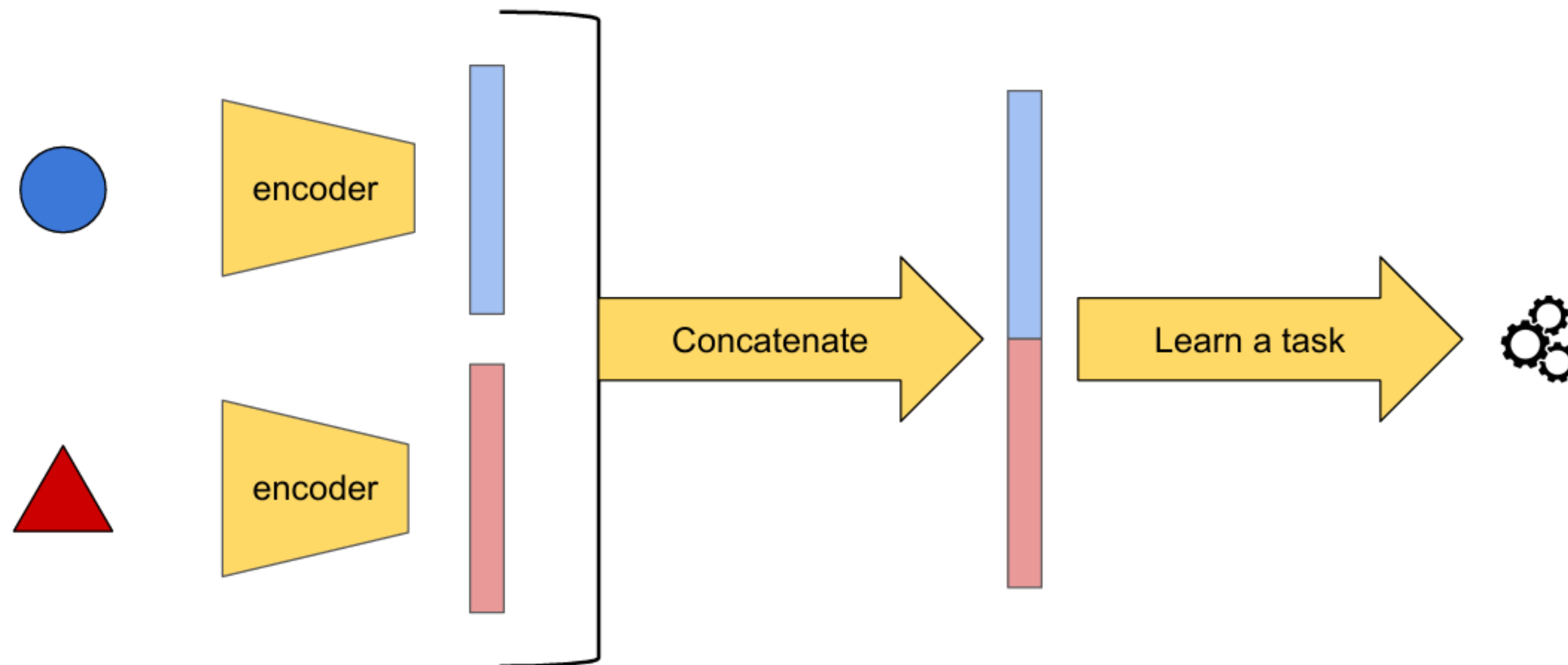
- Bilinear:  $\mathbf{u}W\mathbf{v}$
- Bilinear gated:  $\mathbf{u}W\sigma(\mathbf{v})$
- Low-rank bilinear:  $\mathbf{u}U^T V\mathbf{v} = P(U\mathbf{u} \odot V\mathbf{v})$
- Compact bilinear:  $\text{FFT}^{-1}(\text{FFT}(\Psi(\mathbf{x}, \mathbf{h}_1, \mathbf{s}_1)) \odot \text{FFT}(\Psi(\mathbf{x}, \mathbf{h}_2, \mathbf{s}_2)))$

create own multimodal dataset



## Training Methodologies

- ▶ **Channels Concatenation**
- ▶ **Feature Concatenation**
- ▶ **Feature Fusion by Addition or Dot Product**
- ▶ **Feature Fusion by Neural Network**

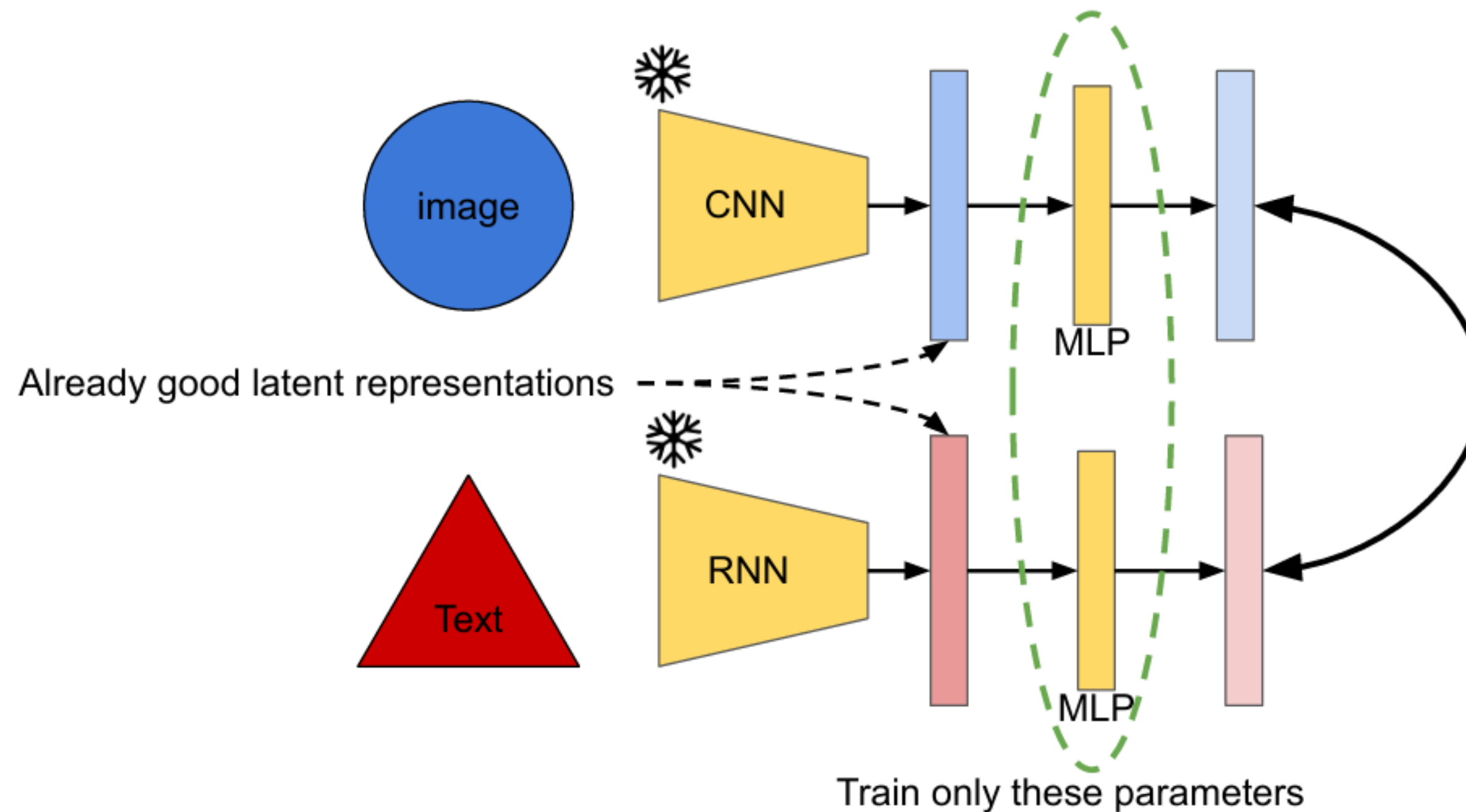


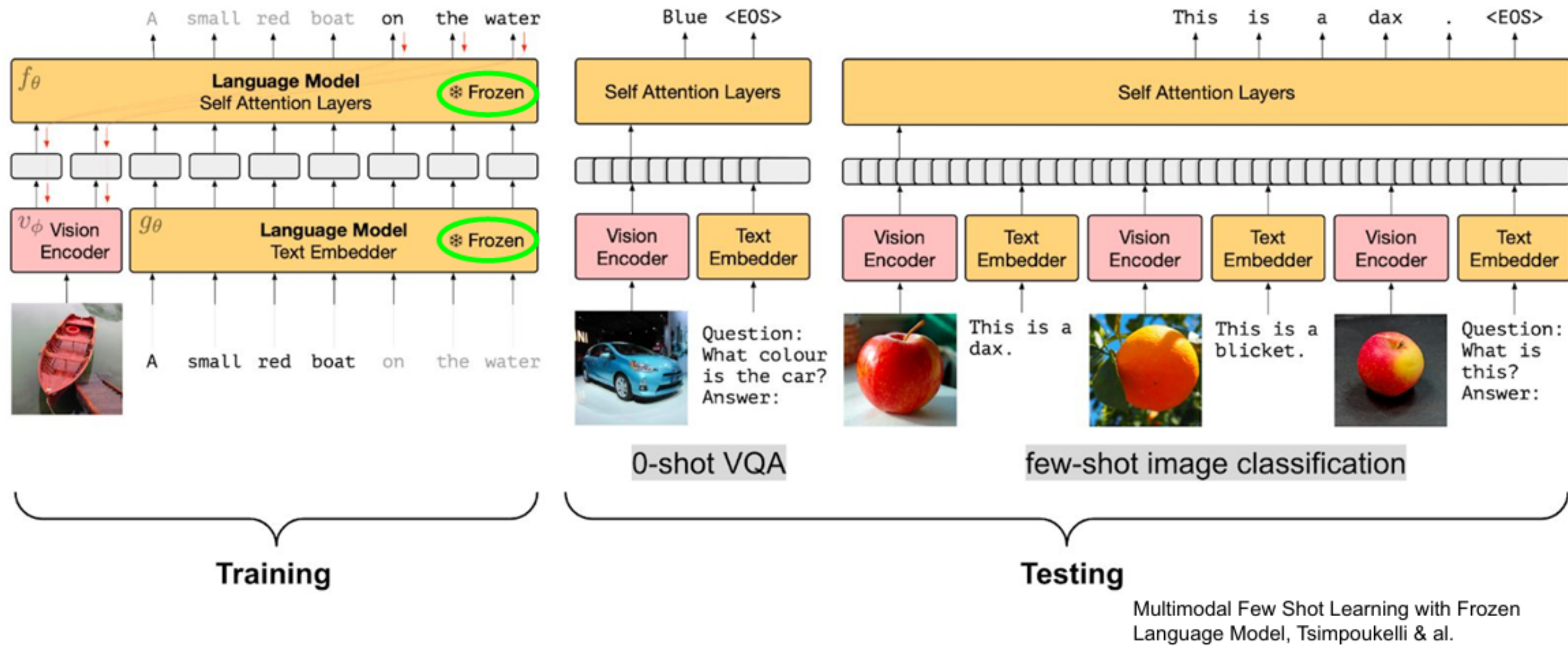


## Training Methodologies

- ▶ Most common approach is with pretrained models
- ▶ We take representations from both modalities from a model that is trained only the one and combine them

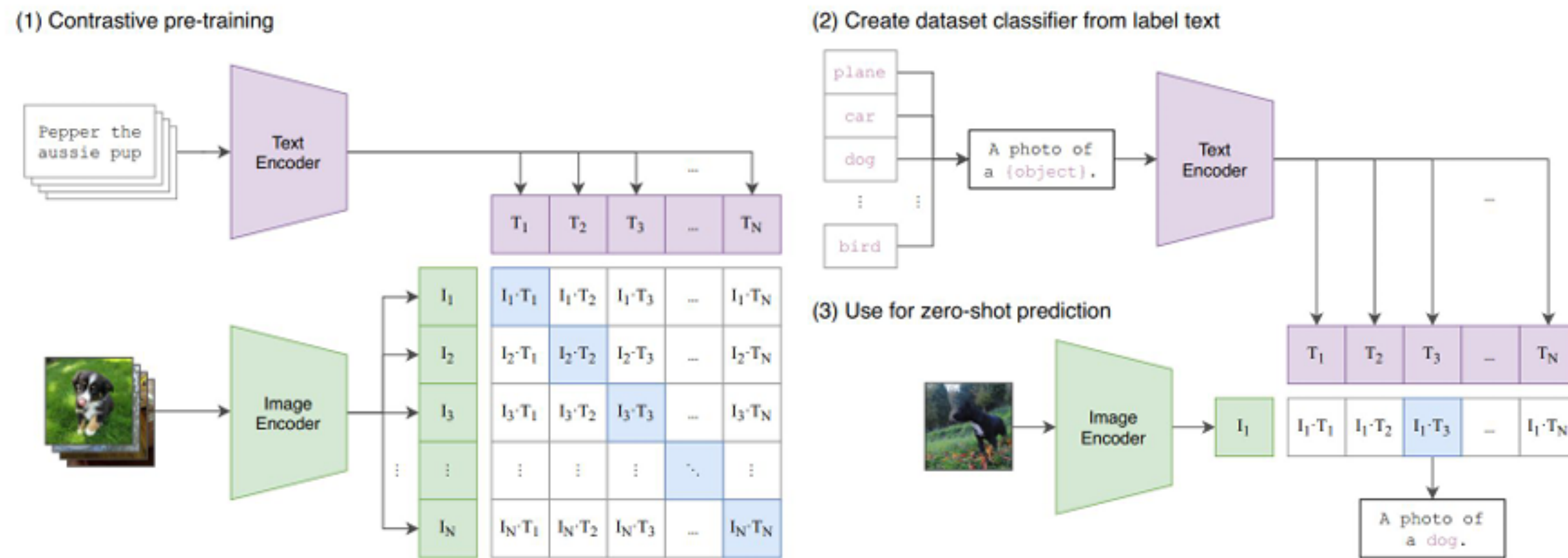
Use pre-trained models to encode different modalities :





## CLIP Model

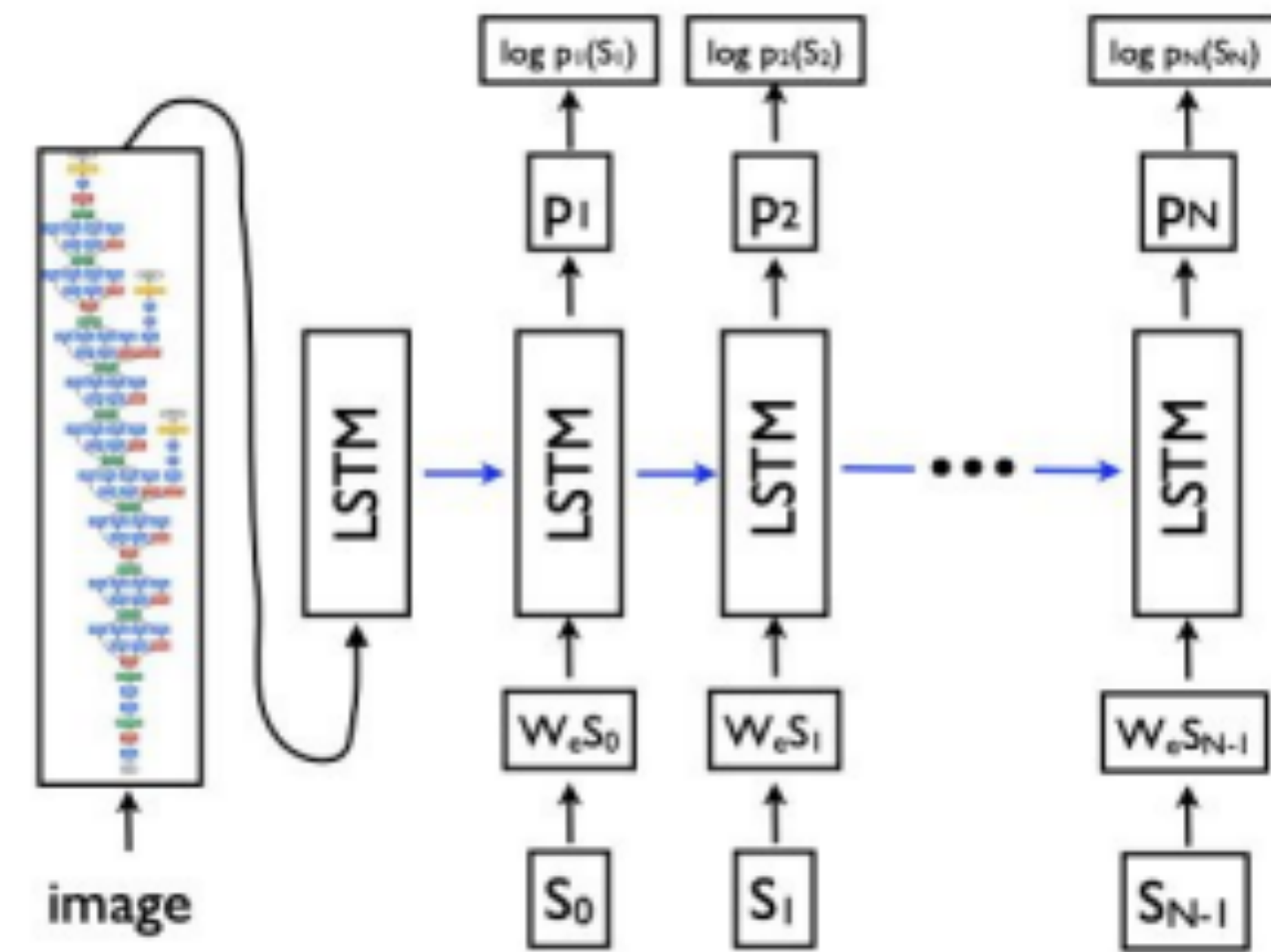
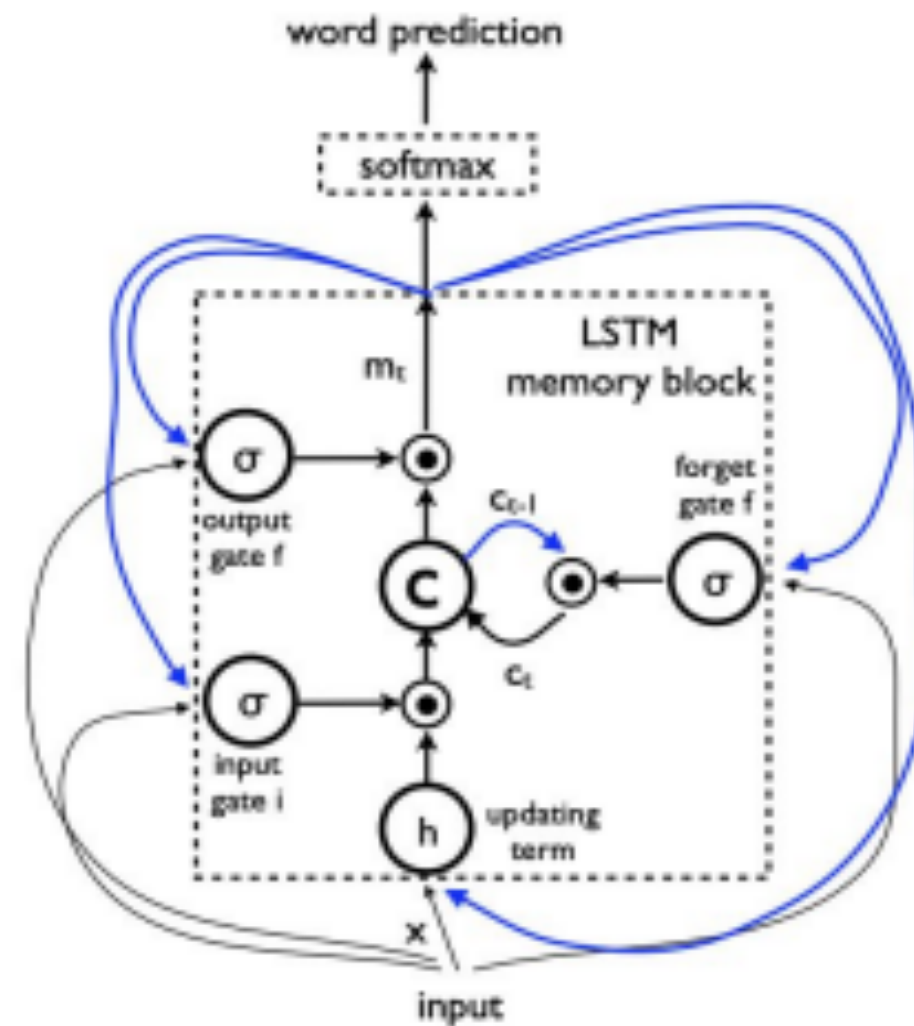
- ▶ **CLIP is one of the most well-known multimodal models**
- ▶ **It utilizes contrastive pre-training to bring the text and image representations close to each other**



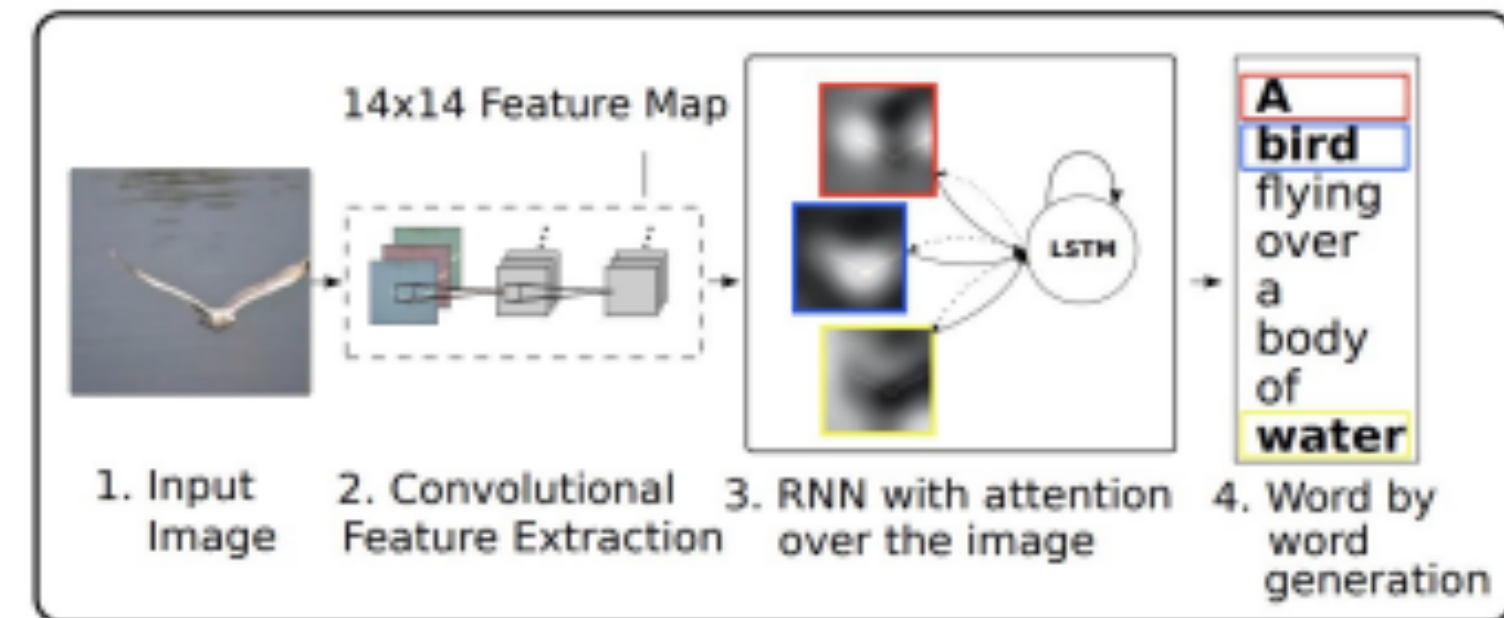
# Image to text: Captioning

Show and tell (Vinyals et al., 2015)

Show, attend and tell (Xu et al., 2016)



Vinyals et al., 2015

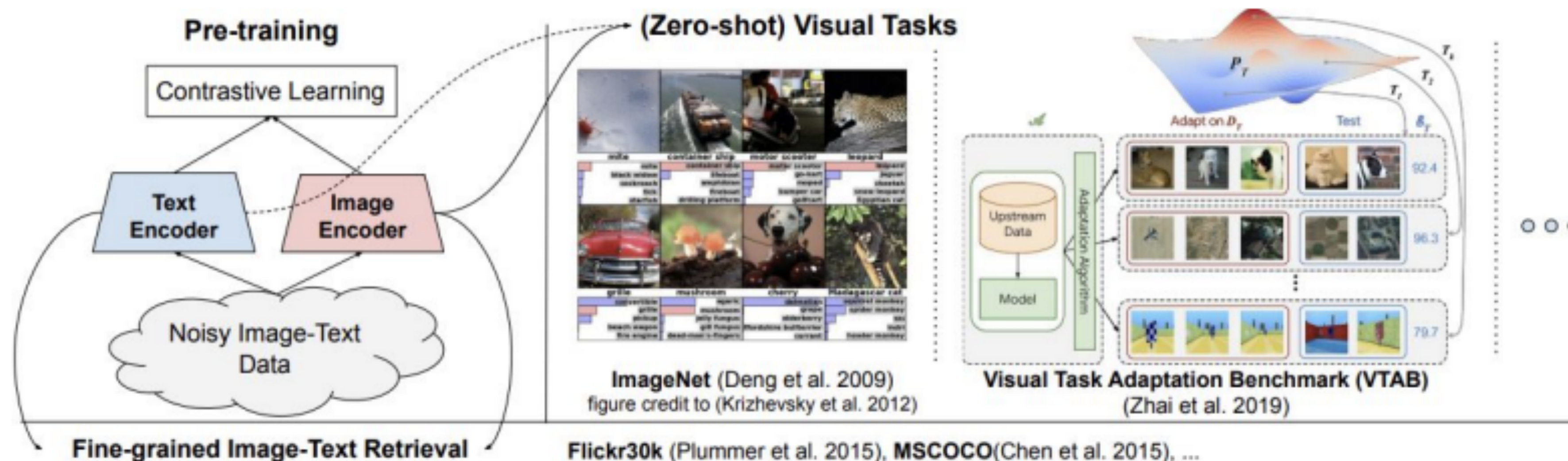


Xu et al., 2016



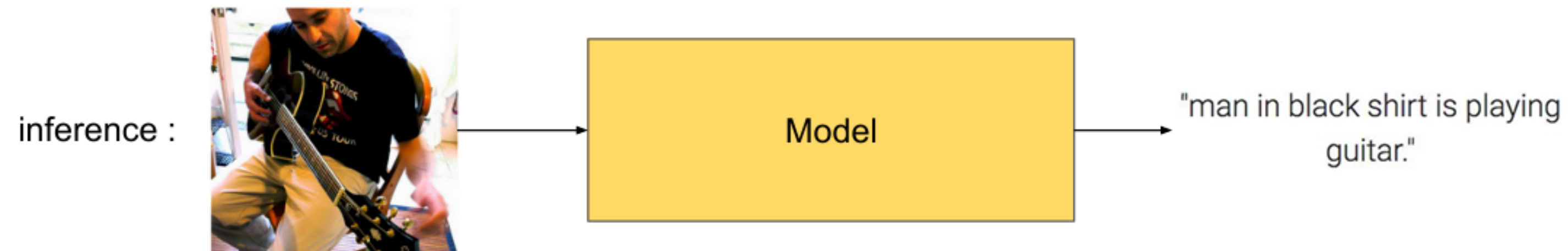
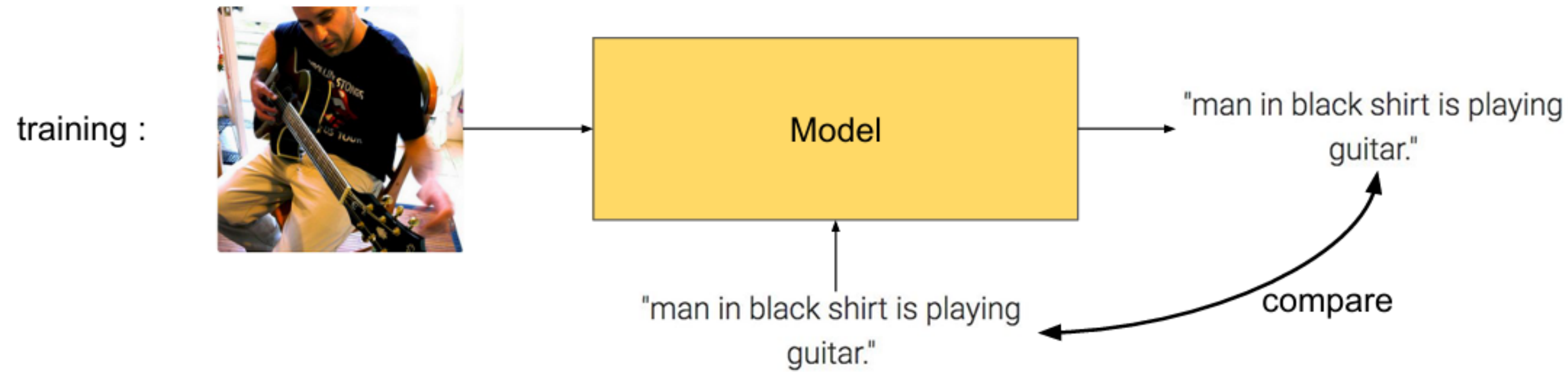
# ALIGN (Jia et al., 2021)

Same idea, but EVEN MORE data (JFT at 1.8B image-text pairs vs CLIP's 300m).



## Well-Known Multimodal Tasks

- ▶ **Image Captioning**
- ▶ **Visual Question Answering**
- ▶ **Multimodal Conversational Systems**
- ▶ **Vision Language Navigation**
- ▶ **Deepfake Detection**





## VQA - Visual Question Answering (Antol et al., 2015)

- The dominant task in vision and language.
  - VQA/VQAv2 citations: 4305+1684
  - COCO Captions: 1647
  - Flickr30k: 1228
- At first the “V” in VQA was found to not matter all that much, so a follow-up VQAv2 dataset was created (Goyal et al., 2017).
- There’s also GQA (Hudson & Manning, 2019).



What color are her eyes?  
What is the mustache made of?



How many slices of pizza are there?  
Is this a vegetarian pizza?



Is this person expecting company?  
What is just under the tree?



Does it appear to be rainy?  
Does this person have 20/20 vision?



Que: What kind of plants are these?  
Ans: flowers



Que: Is the kite high in the air?  
Ans: yes



Que: Where is the light coming from?  
Ans: window





What is this bird?



This is a male song sparrow (*Passer domesticus*) in a feeder. The bird is eating seeds.

What other birds can this feeder attract?



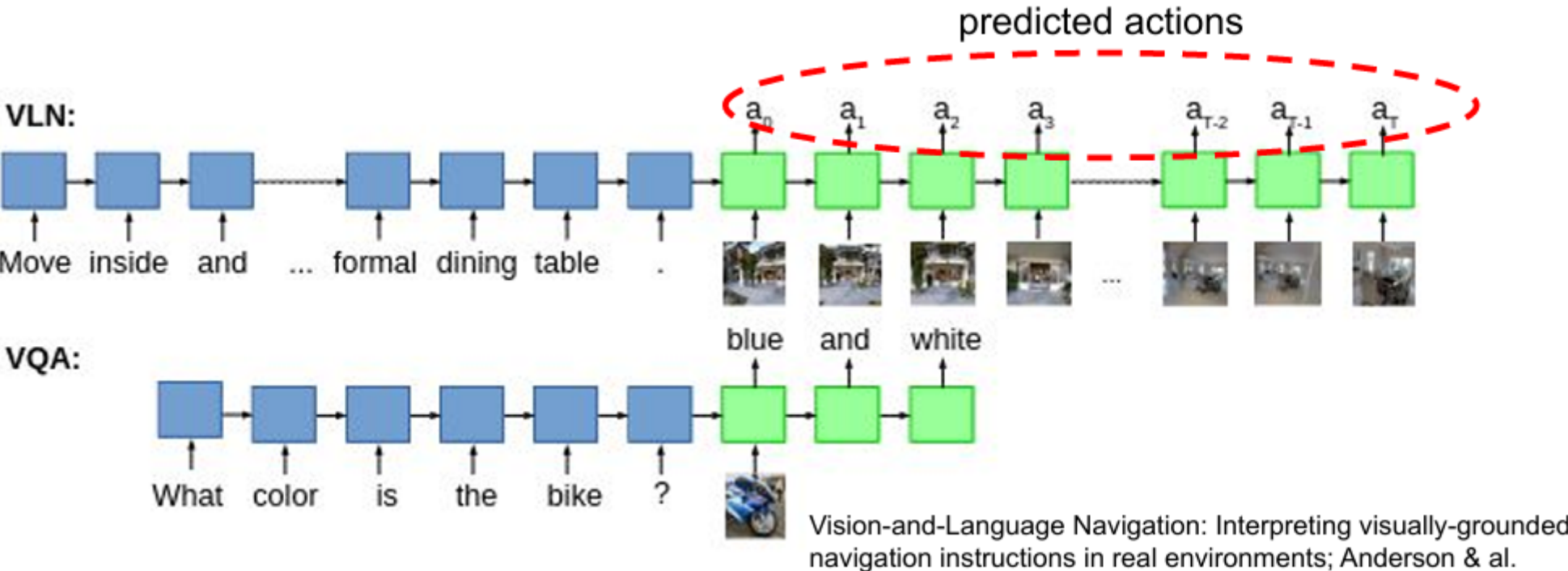
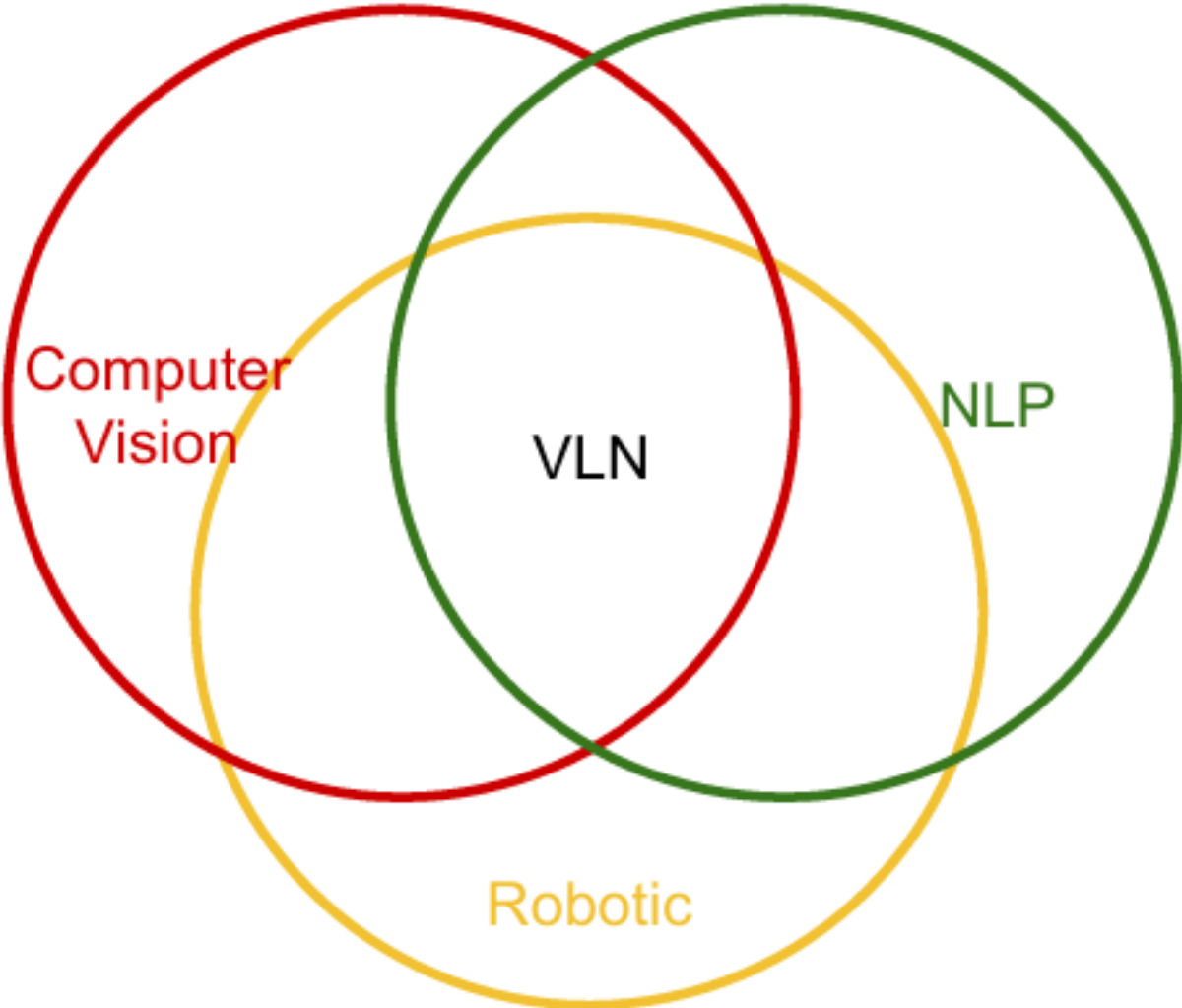
This feeder can attract many types of birds. The birds that can eat the seeds include the song sparrow, the bluebird, the titmouse, the black-headed hummingbird and the goldfinch.

Show me pictures of some of these birds.

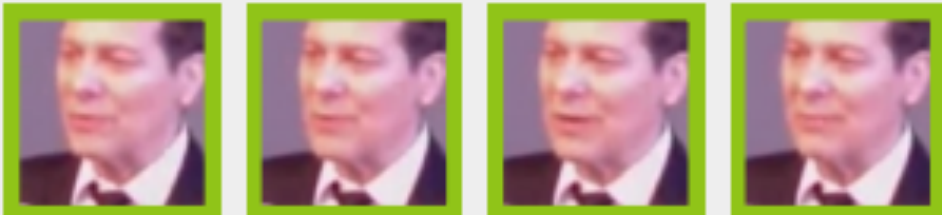
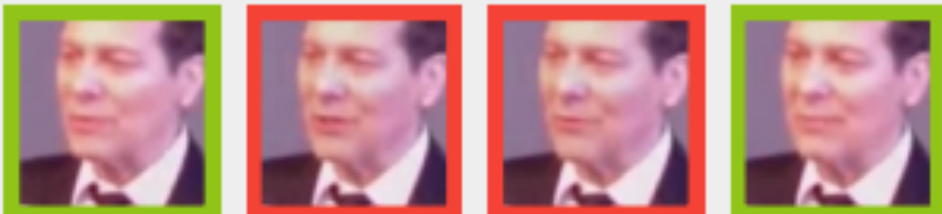
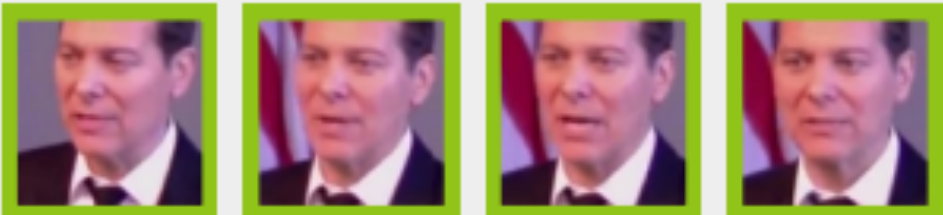
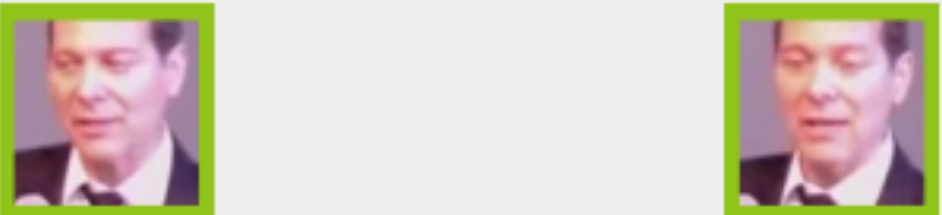
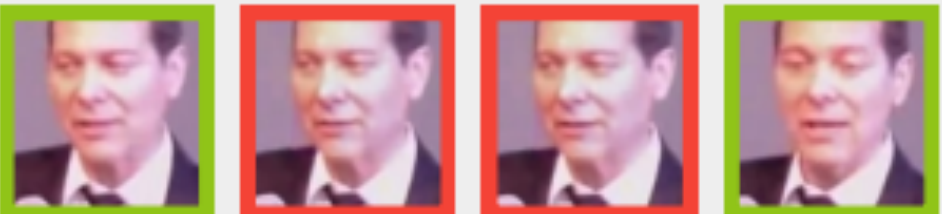


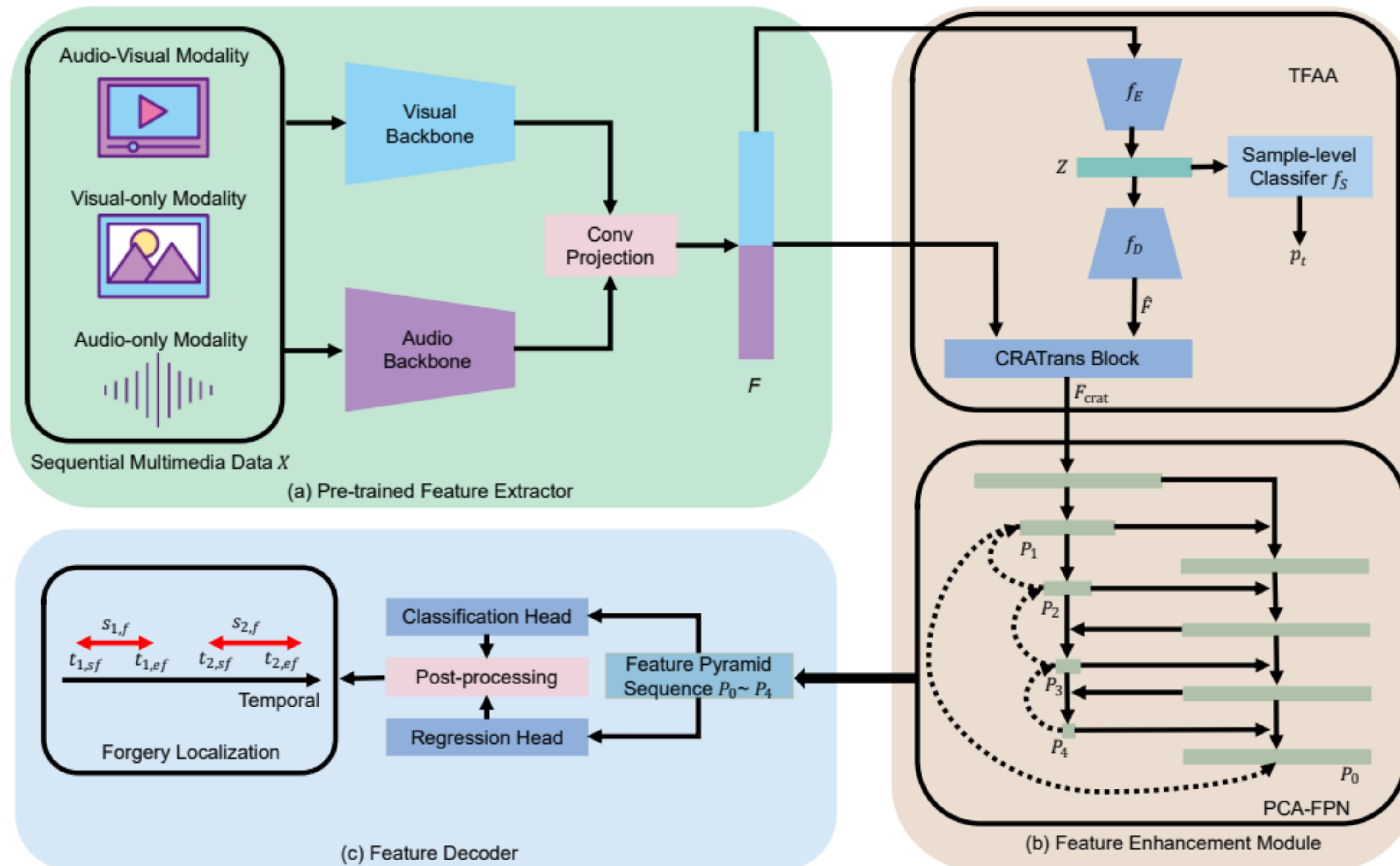
Here are pictures of some of the birds that can eat the seeds that this feeder attracts.







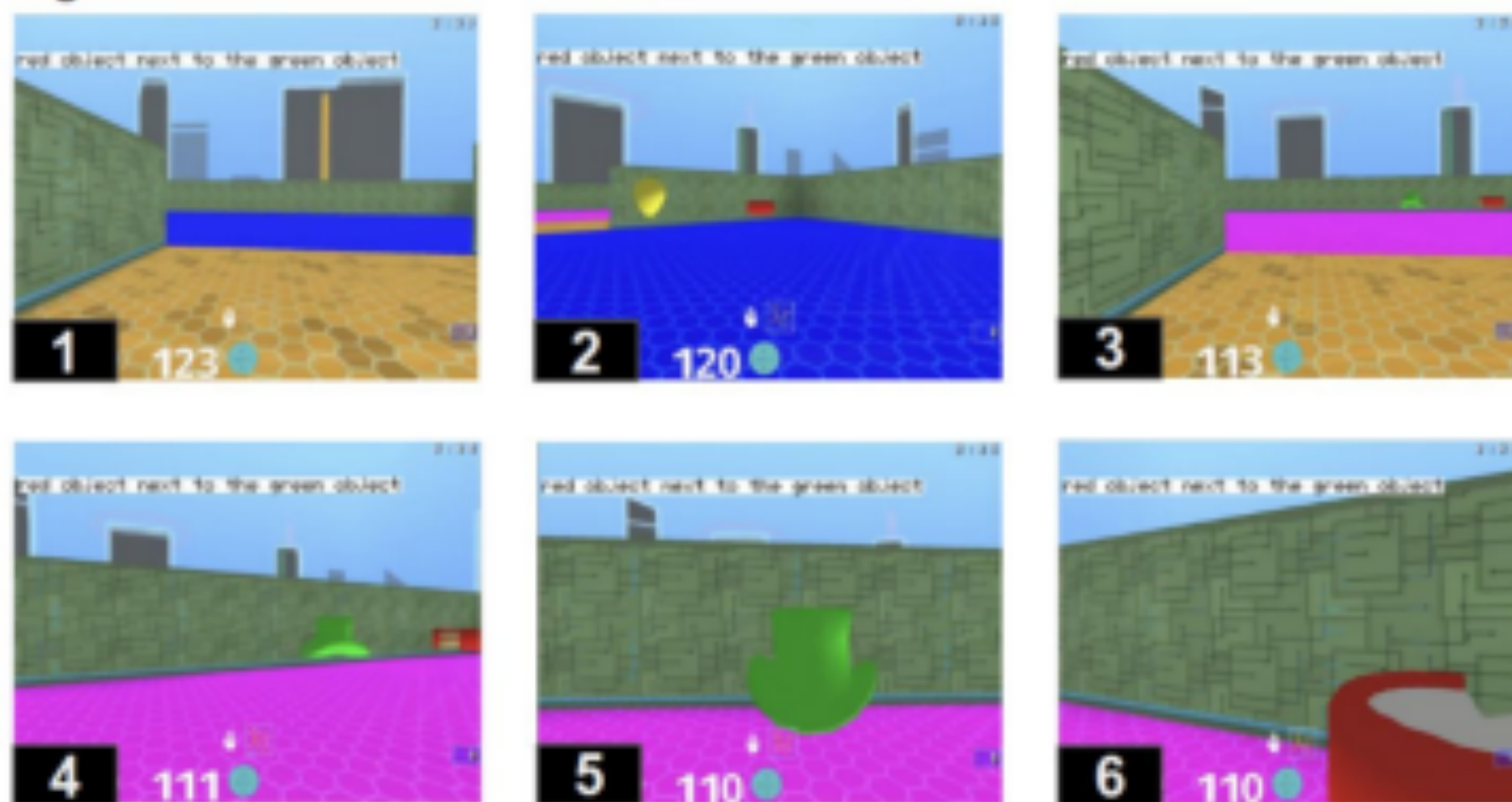
|                       | Real                                                                                | Fake                                                                                  |
|-----------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| the great songbook    |   |    |
| I'm not going to      |  |   |
| and unique            |  |   |
| the terrible songbook |   |   |
| I'm not going to      |  |  |
| and not unique        |  |  |



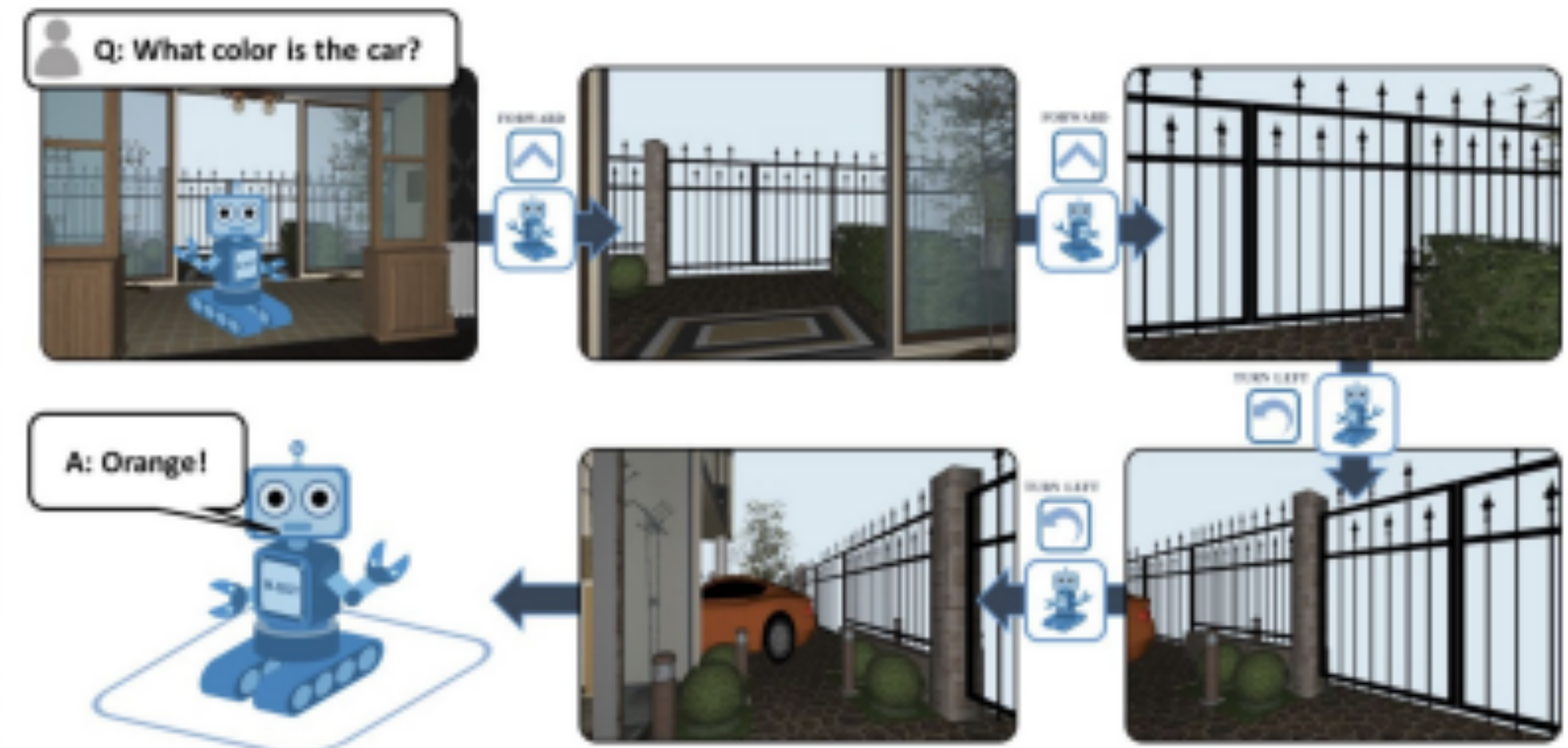


## Grounded language learning in simulated environments

Agent view



Hermann, Hill, et al. 2017



Das et al., 2018



# Text to 3D

POINT-E (Nichol, Jun, et al., 2022)

